

MANIWAKI, QC, Canada

WMO: 717210

Lat: 46.2736N

Lon: 75.9878W

Elev: 655

StdP: 14.35

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.7	-13.1	-26.5	1.3	-17.4	-21.0	1.8	-11.4	17.2	23.2	15.5	21.8	2.7	330	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	85.8	68.6	82.6	67.1	79.7	65.8	72.5	80.2	70.6	77.8	68.9	75.5	8.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.2	113.8	75.5	68.3	106.6	73.9	66.5	100.0	72.3	36.6	80.3	35.0	77.8	33.5	75.2	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	13.4	11.7	-26.1	90.7	5.4	2.9	-30.0	92.8	-33.2	94.6	-36.2	96.2	-40.2	98.3
			-26.4	75.9	5.3	2.5	-30.2	77.7	-33.2	79.1	-36.2	80.5	-40.0	82.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.0	10.7	14.3	25.4	39.6	53.6	62.6	66.6	64.6	57.0	44.8	32.0	18.8	(6)
	DBStd	21.67	13.84	12.01	11.56	8.80	8.20	6.91	5.29	5.70	7.71	8.19	9.57	12.11	(7)
	HDD50	5103	1219	999	764	330	54	2	0	0	21	207	541	966	(8)
	HDD65	9013	1684	1419	1227	763	366	124	43	79	260	627	990	1431	(9)
	CDD50	1812	0	0	2	18	165	381	516	453	231	45	1	0	(10)
	CDD65	245	0	0	0	0	12	53	93	67	19	1	0	0	(11)
	CDH74	2397	0	0	1	14	196	582	867	557	172	8	0	0	(12)
	CDH80	596	0	0	0	2	47	164	226	118	39	0	0	0	(13)
(14) Wind	WSAvg	5.0	4.9	5.1	5.5	5.9	5.4	4.8	4.5	4.3	4.5	5.1	5.2	4.7	(14)
(15) Precipitation	PrecAvg	35.6	2.6	1.9	2.4	2.4	2.7	3.5	3.6	3.7	3.6	3.1	3.0	2.8	(15)
	PrecMax	43.3	5.1	4.6	5.6	4.2	4.6	5.8	6.9	7.5	7.1	5.2	7.1	5.0	(16)
	PrecMin	28.9	1.0	0.7	0.6	0.8	1.2	1.6	1.5	0.8	1.1	1.5	0.6	1.0	(17)
	PrecStd	4.1	1.0	0.9	1.3	0.9	1.1	1.2	1.7	1.4	1.6	0.9	1.3	1.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	44.9	60.6	75.4	85.6	89.4	90.2	87.9	84.9	73.5	59.3	47.4	(19)
		MCWB	44.9	39.5	49.7	57.1	64.2	69.6	71.8	70.0	69.3	61.4	54.5	44.8	(20)
	2%	DB	38.8	39.3	50.5	66.4	79.5	84.5	85.4	83.2	78.7	67.1	53.8	41.3	(21)
		MCWB	37.5	35.8	42.3	50.2	60.5	67.4	69.2	68.3	67.8	58.3	50.2	39.5	(22)
	5%	DB	34.5	35.7	45.4	60.2	74.4	80.7	82.0	79.9	74.4	62.4	49.6	37.5	(23)
		MCWB	33.2	33.1	38.5	48.0	57.5	65.7	67.4	66.7	64.5	55.5	46.4	35.8	(24)
	10%	DB	30.9	32.4	41.1	54.6	69.3	76.6	79.0	76.8	70.6	58.0	45.4	34.4	(25)
		MCWB	29.5	30.5	35.9	44.3	56.1	63.3	66.2	65.2	63.4	52.8	42.2	32.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.9	40.7	52.5	60.4	69.3	74.4	75.5	73.7	72.3	65.2	55.9	45.7	(27)
		MCDB	45.4	42.4	58.7	69.0	79.9	83.3	84.5	80.9	79.4	68.5	57.7	46.6	(28)
	2%	WB	37.8	36.5	43.8	54.6	64.8	71.1	72.8	71.4	69.4	60.0	50.8	40.0	(29)
		MCDB	38.6	38.6	48.1	62.6	73.7	79.9	80.4	77.9	75.1	66.0	53.3	41.1	(30)
	5%	WB	33.3	33.4	39.6	49.1	61.5	68.7	70.7	69.5	67.0	56.6	46.7	36.1	(31)
		MCDB	34.3	35.4	44.3	57.3	69.2	76.1	77.6	75.8	72.5	61.1	49.0	37.4	(32)
	10%	WB	29.5	30.6	36.4	45.5	58.3	66.2	68.7	67.5	64.2	53.3	42.6	33.2	(33)
		MCDB	30.7	32.7	40.4	53.0	66.0	73.3	75.2	73.8	69.2	57.6	45.2	34.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	18.0	19.8	19.5	19.9	22.1	21.3	21.0	20.7	20.3	16.5	13.9	14.7	(35)
		MCDBR	15.9	18.9	22.4	31.1	30.3	27.1	25.3	24.7	24.9	23.4	19.3	13.6	(36)
	5% WB	MCWBR	15.4	16.3	14.9	17.7	13.9	12.3	11.8	11.8	14.1	15.0	16.0	12.9	(37)
		MCWBR	15.7	16.6	20.3	26.0	23.4	21.9	20.8	20.0	21.4	20.5	18.1	13.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.262	0.281	0.316	0.341	0.371	0.396	0.396	0.383	0.365	0.334	0.302	0.269	(40)
		taud	2.329	2.253	2.282	2.403	2.377	2.329	2.333	2.383	2.414	2.502	2.501	2.366	(41)
	Ebn at Noon		268	284	289	291	284	276	274	274	269	263	252	252	(42)
		Edn at Noon	23	31	36	35	37	39	38	35	31	25	20	21	(43)
(44) All-Sky Solar Radiation	RadAvg		422	727	1171	1460	1625	1787	1765	1548	1198	699	409	306	(44)
	RadStd		64	96	110	123	150	127	157	83	112	73	50	31	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (18 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page